Authoritative technical references

Research update: **2026-11-06, America/Toronto, T6**. Primary guidance supports the focused four-week specification; it does not establish installed versions or executed commands. Recheck compatibility, advisories, lawful terms and installed help at stage entry. No software/provider subscription is purchased by T6.

1. Standards and namespaces

- NIST SP 800-207: resource-focused principles; no compliance certification.
- RFC 2606: .test is reserved for testing.
- Earlier detailed NIST/CIS mappings remain historical context in Git; they are not additional evidence requirements.

2. Proxmox

- Official documentation and administration guide: select the installed release.
- Official network source and installation source: bridges and installer/disk cautions.
- Direct admin-guide HTML returned 403 during T6 research; official source/download references were used. No installed-backend or disk command was verified on a host.

3. OPNsense

Project 01 uses bare-metal OPNsense on a separate HP desktop. Use the prepared infrastructure reference to check how it connects to the switch and Dell host.

- Installation and initial configuration and hardware sizing support the separate HP assessment.
- Firewall processing: incoming/stateful policy and automatic/floating/group/interface order; quick and non-quick behavior differ.
- Firewall logging: log/counter interpretation and existing-state limits.
- Interfaces: exact NIC assignment, private/bogon settings.
- Virtual installation supports the superseded design recorded in the decision register.

4. Windows evaluation and hardware

- Windows Server Evaluation and hardware requirements: check lawful terms, activation/expiry and selected version requirements.
- Dell Precision 5820 manuals: match exact chassis, firmware, cooler, processor, storage/adapter and slot before approval. T6 does not certify the ordered parts' fit.
- The latest owner procurement plan/receipt records remain private. Public documents record order status and expected dates only, not serials, order IDs or real inventory addresses.

5. Readiness and network commands

- Get-CimInstance and Get-NetIPConfiguration: proposed read-only local readiness examples; outputs stay private.

6. Active Directory, DNS, and policy

- AD DS deployment, Install-WindowsFeature and Install-ADDSForest: lab-only installation, secure prompt and reboot considerations.
- AD firewall requirements: scope required services to approved lab systems. Same-zone host rules still matter.

7. Endpoint and certificate verification

- Test-ComputerSecureChannel: supported member-machine check, not DC health. Initial direct fetch failed; Microsoft indexed documentation confirmed syntax. Verify locally installed help before use.
- OpenSSL req, x509 and verify: one CA, one leaf, extensions, chain/name/date checks. Commands target OpenSSL 3.x and must match installed help.
- Apache TLS module: certificate/key and protocol directives. No AD CS/separate issuer or certificate-revocation infrastructure is required.

8. Application authentication and authorization

- Apache authentication guide, file groups, authorization merge behavior and htpasswd: local hashed accounts, group-protected directories, AuthMerging and interactive passwords.
- Apache control wrapper: configtest; package activation/path wrappers differ.
- File-based roles are manually mapped to AD groups; these references do not establish AD SSO or synchronization.

9. Tailscale

- Access control, grant syntax and grant examples: explicit additive policy, exact destination and TCP 443.
- ACL examples: review permissive initial/default policy; omitted content is not proof of deny-all.
- CLI reference: proposed status inspection, private output.
- Device removal: verify current revoke/remove behavior and actual existing/new session denial.
- Recheck current plan/MFA/logging/identity terms at stage entry. No production tailnet, subnet routing, exit node, public sharing or remote management service is included.

10. Twingate

Historical navigation anchor only. Twingate is excluded from the T6 baseline; no second provider, connector or comparison is required. Earlier references remain in Git history.

11. Logging and automation

- Kerberos authentication auditing: confirm which host emits the actual directory authentication events.
- systemd journalctl: read-only --since/--no-pager examples. Apache request logs are separate.
- Existing PowerShell/Python supports one local capacity-summary operation. T6 adds no executable lab automation, Ansible requirement, collector VM or SIEM.

12. GitHub and repository workflow

- Actions security hardening and official checkout: least privilege, full commit pin, no retained checkout credentials.
- GitHub CLI authentication: existing secure authentication; never expose tokens.
- Current build instructions define offline document/PDF checks.

13. Command verification limits

T6 reviewed primary vendor descriptions/syntax for proposed inventory, AD deployment, member checks, OpenSSL, Apache, Tailscale and journal examples. It executed none on lab assets. Installed versions/backends, actual paths, compatibility, activation, provider behavior, recovery targets and Linux package commands require a future approved change record. Documentation CI fetches no external references and reaches no lab systems.